Adjoints

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We will be using ideas from the discussion on ***Free constructions****. Use the following examples for thinking concretely: free groups made from sets, and vector spaces as free constructions on their basis sets.*

Consider the forgetful functor U (for "underlying") from $\boldsymbol{D} = \mathbf{Grp}$ to $\boldsymbol{C} = \mathbf{Set}$. We want to create a functor $F : \boldsymbol{C} \to \boldsymbol{D}$ such that the object map of this functor is the "create free construction" map, i.e., for an arbitrary set $X \in \text{obj}(\boldsymbol{C})$, $F(X)$ would be a free group based on X.

Observe that in order for free constructions to exist, we must have universal morphisms from every $X \in \text{obj}(\boldsymbol{C})$ to the forgetful functor U (Since we cannot just take it for granted that $!_X \downarrow U$ will have initial objects). For an arbitrary X, let us denote one of the universal morphisms from X to U as $\left(\hat{X}, i_X : X \to U\left(\hat{X}\right)\right)$.

Now, given such a functor U, we can define the aforementioned F, which we will call the left-adjoint of U.

We define $F_{\text{obj}} : X \mapsto \hat{X}$.

Observe that for $c \in \text{obj}(\boldsymbol{C}), d \in \text{obj}(\boldsymbol{D})$, every morphism from c to $U(d)$ uniquely defines a morphism from $\hat{c}$ to d (by virtue of $(\hat{c}, i_c)$ being an universal morphism from c to U). In the inverse direction, every morphism f from $\hat{c}$ to d produces a map from c to $U(d)$, namely $U(f) \circ i_c$. Therefore, we have

$$\text{Hom}_C(c, U(d)) \cong \text{Hom}_D(F(c), d)$$

This is why F is called the *left*-adjoint of U.

The name "adjoint" is probably inspired by the equation

$$\langle x, Ty \rangle = \langle T^* x, y \rangle,$$

where T^* is the complex conjugate of T, also known as the *adjoint* of T.

Given any $f \in \text{Hom}_C(X, Y)$, we use the bijection $\text{Hom}_C(X, U(F(Y))) \cong \text{Hom}_D(F(X), F(Y))$ on $i_Y \circ f$ to produce a map from $F(X)$ to $F(Y)$. We define this map to be $F_{\text{mor}}(f)$.

Therefore, $F(f)$ is the unique morphism which makes the following square commute:

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ \Big\downarrow{\scriptstyle i_X} & & \Big\downarrow{\scriptstyle i_Y} \\ U(F(X)) & \xrightarrow[U(F(f))]{} & U(F(Y)) \end{array}$$

This shows that there is a natural transformation from id_C to $U \circ F$, with $\{i_A\}_{A \in C}$ as its components. This natural transformation is known as the "unit". Try again later. : Relate with unit-counit definition.

Observe that the morphism-map for F just fell out naturally after we had the fact that

$$\mathrm{Hom}_D(F(c), d) \cong \mathrm{Hom}_C(c, U(d))$$

We can also recover i_X from this isomorphism by pulling $\mathrm{id}_{F(X)}$ through the isomorphism.

What happens if we start just with the isomorphism as our only assumption? But what kind of isomorphism do we want between $\mathrm{Hom}_D(F(c), d)$ and $\mathrm{Hom}_C(c, U(d))$?

Suppose we have set bijections $\varphi_{c,d} : \mathrm{Hom}_D(F(c), d) \to \mathrm{Hom}_C(c, U(d))$. For them to behave nicely with the categories C and D (with respect to composition), we should have that the following squares commute :

$$\begin{array}{ccc} \mathrm{Hom}_D(F(c), d) & \xleftrightarrow{\varphi_{c,d}} & \mathrm{Hom}_C(c, U(d)) \\ \downarrow{\scriptstyle g \circ -} & & \downarrow{\scriptstyle U(g) \circ -} \\ \mathrm{Hom}_D(F(c), d') & \xleftrightarrow[\varphi_{c,d'}]{} & \mathrm{Hom}_C(c, U(d')) \end{array} \qquad \begin{array}{ccc} \mathrm{Hom}_D(F(c), d) & \xleftrightarrow{\varphi_{c,d}} & \mathrm{Hom}_C(c, U(d)) \\ \downarrow{\scriptstyle - \circ F(h)} & & \downarrow{\scriptstyle - \circ h} \\ \mathrm{Hom}_D(F(c'), d) & \xleftrightarrow[\varphi_{c',d}]{} & \mathrm{Hom}_C(c', U(d)) \end{array}$$

where $g \in \mathrm{Hom}_D(d, d')$ and $h \in \mathrm{Hom}_C(c', c)$. Note that the φ arrows are bidirectional.

To motivate the forward-φ direction of the first commutative square, draw out categories C and D with the objects $c, U(d), U(d')$ and $F(c), d, d'$ respectively.

Now take $f \in \mathrm{Hom}_D(F(c), d), g \in \mathrm{Hom}_D(d, d')$. Observe that we need (for "aesthetic" reasons)

$$\varphi_{c,d'}(g \circ f) = U(g) \circ \varphi_{c,d}(f),$$

which is precisely what the square claims.

Similarly, handle the motivation of the backward-φ direction of the first square and both φ directions of the second square.

They correspond respectively to $\mathrm{Hom}_D(F(c), -), \mathrm{Hom}_C(c, U(-)) : D \to \mathbf{Set}$ being naturally isomorphic and $\mathrm{Hom}_D(F(-), d), \mathrm{Hom}(-, U(d)) : C^{\mathrm{op}} \to \mathbf{Set}$ being naturally isomorphic.

Putting those two together, we get that the two bifunctors $\mathrm{Hom}_D(F(-), -), \mathrm{Hom}_C(-, U(-)) : C^{\mathrm{op}} \times D \to \mathbf{Set}$ should be naturally isomorphic.

Therefore, we finally get that $F : C \to D$ and $U : D \to C$ are called adjoints of each other iff

$$\mathrm{Hom}_D(F(-), -), \mathrm{Hom}_C(-, U(-)) : C^{\mathrm{op}} \times D \to \mathbf{Set}$$

are naturally isomorphic functors.

Rephrasing, functors $F : C \to D$ and $U : D \to C$ are called adjoints of each other iff the isomorphism

$$\mathrm{Hom}_D(F(c), d) \cong \mathrm{Hom}_C(c, U(d))$$

is natural in the arguments c and d.

Observe that the key part was

$$\forall c, d, \mathrm{Hom}_D(F(-), -) \cong \mathrm{Hom}_C(-, U(-)) \text{ as sets.}$$

The isomorphism being natural in c and d is merely book-keeping to make sure that the set theoretic isomorphism plays well along with the composition in the groups C and D.

This might be a recurring theme about naturality, in the sense that it tries to make sure that the basic structure of categories is being preserved. I also feel like whenever naturality is used somewhere in a definition, it is just there to make sure that the structure is being maintained, while the real action takes part somewhere else in the definition; but I am not sure about this.

The idea of using $\mathrm{Hom}_D(F(c), d) \cong \mathrm{Hom}_C(c, U(d))$ fits in nicely with

- The product bifunctor is the right adjoint of the diagonal functor:

$$\mathrm{Hom}_{\mathbf{Set} \times \mathbf{Set}}(\langle C, C\rangle, \langle A, B\rangle) \cong \mathrm{Hom}_{\mathbf{Set}}(C, A \times B)$$

- The coproduct bifunctor is the left adjoint of the diagonal functor:

$$\mathrm{Hom}_{\mathbf{Set}}(A \sqcup B, C) \cong \mathrm{Hom}_{\mathbf{Set} \times \mathbf{Set}}(\langle A, B\rangle, \langle C, C\rangle)$$

- Ceiling is the left adjoint of the inclusion functor $i : \mathbb{Z} \to \mathbb{R}$ (considering posetal categories $\mathbb{Z}$ and $\mathbb{R}$):

$$\mathrm{Hom}_{\mathbb{Z}}(\lceil a \rceil, b) \cong \mathrm{Hom}_{\mathbb{R}}(a, i(b))$$

- Floor is the right adjoint of the inclusion functor $i : \mathbb{Z} \to \mathbb{R}$ (considering posetal categories $\mathbb{Z}$ and $\mathbb{R}$):

$$\mathrm{Hom}_{\mathbb{R}}(i(a), b) \cong \mathrm{Hom}_{\mathbb{Z}}(a, \lfloor b \rfloor)$$

- Free group construction is the left adjoint of the forgetful functor from $\mathbf{Grp}$ to $\mathbf{Set}$:

$$\mathrm{Hom}_{\mathbf{Grp}}(\mathrm{Free}(X), G) \cong \mathrm{Hom}_{\mathbf{Set}}(X, \mathrm{Forget}(G))$$

- $(- \times S)$ is the left adjoint of $\mathrm{Hom}(S, -)$:

$$\mathrm{Hom}_{\mathbf{Set}}(A \times S, B) \cong \mathrm{Hom}_{\mathbf{Set}}(A, \mathrm{Hom}(S, B))$$

- $G \mapsto G^{\mathrm{ab}} = G/[G, G]$ is the left adjoint of the inclusion functor $i : \mathbf{Ab} \to \mathbf{Grp}$:

$$\mathrm{Hom}_{\mathbf{Ab}}(G/[G, G], H) \cong \mathrm{Hom}_{\mathbf{Grp}}(G, i(H))$$

Therefore, observing a "simple" isomorphism between $\mathrm{Hom}_D(F(c), d)$ and $\mathrm{Hom}_C(c, U(d))$ is often a key step to identifying that adjoints are at play.

Try again later. : Work left to do:

- What is this feeling/intuition of the left adjoint being "liberal", and the right adjoint being "conservative"? See the comment here.
- What is the intuition behind adjoints being the "formulaic solution to an optimization problem"? See this MO answer.
- The MO answer linked above also says that

  > *"The notion that F is the most efficient solution to the (optimization) problem posed by G is, in a certain rigorous sense, equivalent to the notion that G poses the most difficult problem which F solves."*

  This might explain the "liberal"/"conservative" intuition described above.

References:

- https://en.wikipedia.org/wiki/Adjoint_functors
- *Categories for the Working Mathematician*, Mac Lane: Introduction, "Adjoints" chapter
- https://mathoverflow.net/q/6551
- https://math.stackexchange.com/q/25455
- https://ncatlab.org/nlab/show/adjoint+functor